

Assignment 3 — Layers Me Todo (MVC + Service)

Concept: Separation of concerns — route / controller / service ka kaam

Level	Medium
Time	1 hour
Kaam	Diya hua code 3 files me todna

Kya seekhoge

- Har layer ka **exactly** kya kaam hai
- Service me **req** / **res** kyun nahi hone chahiye
- Logic alag karne se **testing aur reuse** kaise aasaan hota hai

Diya hua code (sab kuch ek hi file me — galat tarika)

```
// src/routes/review.route.js
const express = require("express");
const ReviewModel = require("../model/reviewModel");
const router = express.Router();

router.post("/createReview", async (req, res) => {
  try {
    const { title, comment, rating, reviewerName } = req.body;

    const alreadyReviewed = await ReviewModel.findOne({ reviewerName, title });
    if (alreadyReviewed) {
      return res.status(400).send("aap ye review pehle de chuke ho");
    }

    const review = await ReviewModel.create({
      title, comment, rating, reviewerName,
    });
    res.send(review);
  } catch (err) {
    console.log(err);
    res.status(500).send("internal server error");
  }
});

router.get("/getReviews", async (req, res) => {
  try {
    const { status, page = 1, limit = 10 } = req.query;
    const filter = {};
    if (status) filter.status = status;

    const reviews = await ReviewModel.find(filter)
      .skip((page - 1) * limit)
      .limit(limit);

    res.send(reviews);
  } catch (err) {
    res.status(500).send("error");
  }
});

module.exports = router;
```

Requirements

Is code ko **3 files** me todo:

1. `src/service/reviewService.js`

- Saara DB logic yahan
- `createReview(data)` aur `getReviews(queryParams)` functions
- **req ya res ek baar bhi nahi aana chahiye**
- Duplicate review mile to `throw` karo (`res.send` mat karo)
- `getReviews` ye return kare: `js { reviews: [...], total: 24, page: 1, totalPages: 3 }` (Hint: `countDocuments` aur `Promise.all`)

2. `src/controller/reviewController.js`

- Har function **maximum 4 line** ka
- `req` se data nikalo, service call karo, response bhejo
- **Sahi status code:** create pe `201`, list pe `200`
- Response ka shape har jagah same: `js { success: true, message: "...", data: ... }`

3. `src/routes/review.route.js`

- Sirf routing. Ek bhi `await` ya `ReviewModel` nahi hona chahiye.

Self-check (submit karne se pehle)

- [] Route file me `ReviewModel` word hai kya? **Nahi hona chahiye**
- [] Controller file me `ReviewModel` word hai kya? **Nahi hona chahiye**
- [] Service file me `req` ya `res` hai kya? **Nahi hona chahiye**
- [] Har controller function 4 line se chhota hai?

Soch ke batao

1. Kal ko tumhe ek **CRON job** likhni hai jo har raat purane review delete kare. Original code (sab kuch route me) se wo logic reuse kar paate? Kyun nahi?
2. Service me `res.status(400).send()` likh doge to kya problem hogi?
3. Duplicate review wala check `400` deta hai. Tumhe kya lagta hai — `400` sahi hai ya `409`? Apna jawab **kyun** ke saath likho.

Bonus

`getReviews` me sorting add karo: `?sortBy=rating:desc` Aur dhyaan rakhna — user koi bhi random field bhej ke DB pe sort na karwa sake (sirf `rating` aur `createdAt` allowed).